PROTECTED 5V LOGIC — EXTERNAL BRANCH FUSE REQUIRED

D1
VIN_FB
ACC_FUSED
STPS2H100AY

D2
VIN_FB
GND
SMBJ16A

C1
VIN_FB
GND
10u 63V EEEFK1J100P

C2
VIN_FB
GND
2.2u 100V X7R

C3
VIN_FB
GND
2.2u 100V X7R

U1
GND
VIN_FB
VIN_EN
VIO33
RON
FB
PGOOD
BST
SW
GND
EN_UVLO
RON
EP_GND
FB
LMS1640QDSSRQ1

PWR_FLAG
GND

PWR_FLAG
ACC_FUSED

PWR_FLAG
VIN_PROT

PWR_FLAG
V5_LOGIC

L1
SW
V5_LOGIC
68u WSS1246T-683MLC

R1
RON
GND
41.2k

R2
VIN_FB
VIO33
1M

R3
VIO33
GND
221k

R4
V5_LOGIC
FB
158k

R5
FB
GND
49.9k

R6
SW
RIPPLE
150k

C4
BST
SW
2.2n 50V X7R

C5
RIPPLE
V5_LOGIC
3.3n 50V X7R

C6
RIPPLE
FB
180p C0G

C7
V5_LOGIC
GND
22u 25V X7R

C8
V5_LOGIC
GND
22u 25V X7R

R7
V5_LOGIC
PGOOD
100k

NOMINAL UNO HEADERS / ISOLATED ILL BREAKOUT

J1
ISO_ILL12_P
ISO_ILL12_N
ISO_ILL11_P
ISO_ILL11_N
GND
ACC_FUSED
CONSOLE_BRANCH_ONLY

JP1
V5_LOGIC
ESP_5V
OPEN: USB ISOLATION

H1
PWM_W
PWM_B
PWM_G
IOL_NOMINAL

H2
ENABLE_REQ
GND
IOH_NOMINAL

H3
ESP_5V
GND
GND
POWER_NOMINAL

H4
PWM_W
AD_NOMINAL

TP1
ISO_ILL1_P

TP2
ISO_ILL1_N

TP3
ISO_ILL2_P

TP4
ISO_ILL2_N

TP5
V5_LOGIC

TP6
GND

TP7
PGOOD

TP8
ENABLE_REQ

MH1
M3 SUPPORT

MH2
M3 SUPPORT

ONE RGBW ZONE — EXTERNALLY POWERED LEDS

J2
LED_W_ACT
LED_GND
LED_B_ACT
R- G- B- W- LED_GND

U2
ENABLE_REQ
PWM_B
GND
PWM_G
DRV_G
V5_LOGIC
DRV_B
ENABLE_REQ
UCC27524AQDRQ1

U3
ENABLE_REQ
PWM_W
GND
PWM_W
DRV_W
V5_LOGIC
DRV_B
ENABLE_REQ
UCC27524AQDRQ1

C9
V5_LOGIC
GND
100n 25V X7R

C11
V5_LOGIC
GND
100n 25V X7R

C10
V5_LOGIC
GND
1u 25V X7R

C12
V5_LOGIC
GND
1u 25V X7R

R8
ENABLE_REQ
GND
2.2k

Q1
GATE_W
LED_W_NEG
BUK9Y15-60E

R9
DRV_W
GATE_W
100

R13
GATE_W
GND
10k

R17
PWM_W
GND
100k

D3
LED_W_ACT
GND
SMBJ24CA

Q2
GATE_G
LED_G_NEG
BUK9Y15-60E

R10
DRV_G
GATE_G
100

R14
GATE_G
GND
10k

R18
PWM_G
GND
100k

D4
LED_W_ACT
GND
SMBJ24CA

Q3
GATE_B
LED_B_NEG
BUK9Y15-60E

R11
DRV_B
GATE_B
100

R15
GATE_B
GND
10k

R19
PWM_B
GND
100k

D5
LED_B_ACT
GND
SMBJ24CA

Q4
GATE_W
LED_W_NEG
BUK9Y15-60E

R12
DRV_W
GATE_W
100

R16
GATE_W
GND
10k

R20
PWM_W
GND
100k

D6
LED_W_ACT
GND
SMBJ24CA

4Runner Vehicle Health OS
Sheet: /
File: vhos_rgbw_rev4.kicad_sch
Title: VHOS RGBW SHIELD — EVALUATION REV A
Size: A1 Date:
KiCad E.D.A. 10.0.6
Rev: A
Id: 1/1

EVALUATION DRAFT / NO VEHICLE CONNECTION RELEASE. JP1 DEFAULT OPEN. M8 & TOYOTA CAVITIES UNVERIFIED.